

NAMIBIA SENIOR SECONDARY CERTIFICATE

BIOLOGY ADVANCED SUBSIDIARY LEVEL

8223/1

PAPER 1 Multiple Choice

1 hour

Marks 40

2023

Additional Materials: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Write in soft pencil.
- Make sure that you receive the multiple choice answer sheet with your examination number on it.
- There are **forty** questions on this paper. Answer **all** questions.
- For each question, there are four possible answers **A, B, C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.
- If you want to change an answer, thoroughly erase the one you wish to delete.

Read the instructions on the answer sheet carefully.

- Each correct answer will score one mark.
- All questions in this paper carry equal marks.
- Any rough working should be done in this booklet.
- You may use a non-programmable calculator.

This document consists of **12** printed pages.



Republic of Namibia
MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 Which row correctly describes the binomial system for naming organisms?

	first part	second part
A	family	genus
B	genus	family
C	genus	species
D	species	family

- 2 Why do researchers use random sampling rather than counting all organisms to determine the biodiversity of an area?

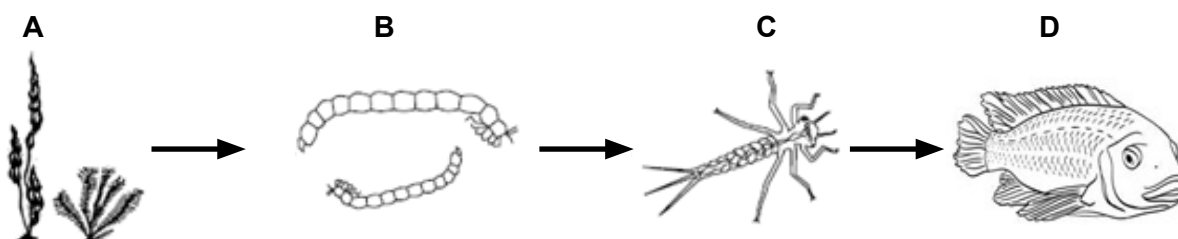
- A** It is easy to get the full list of a specific population.
- B** Not all organisms have equal chances of being selected.
- C** It is considered to be a representative of all organisms.
- D** Some areas are too small and the organisms are too large.

- 3 In a food chain, a tree is called a producer; it produces organic compounds during photosynthesis. Animals obtain organic compounds by feeding on the tree and it also provides shelter to small animals.

Which term applies to the description of the tree?

- A** ecosystem
- B** community
- C** food web
- D** niche

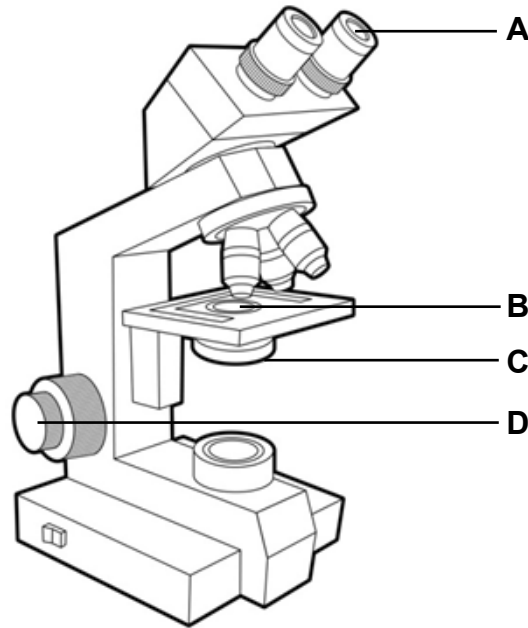
- 4 Bromadiolene is a chemical contained in pesticides that can undergo bioaccumulation when it ends up in water. In which of the organisms in the food chain below would the amount of bromadiolene be highest?



- 5 Why is it important to control the spread of alien plant species?

- A** They do not contain essential nutrients for animals.
- B** They have a potential of replacing indigenous plant species.
- C** They are often very big trees.
- D** They lead to increased energy loss in the food chains.

- 6 The diagram shows a light microscope.



Which part of the microscope magnifies the object to produce the final image?

- 7 The diameters of living cells varies considerably. The smallest free living eukaryote is believed to be $0.84\text{ }\mu\text{m}$ in diameter and the smallest prokaryote is 30 nm .

Which one is greater in diameter and by how many times?

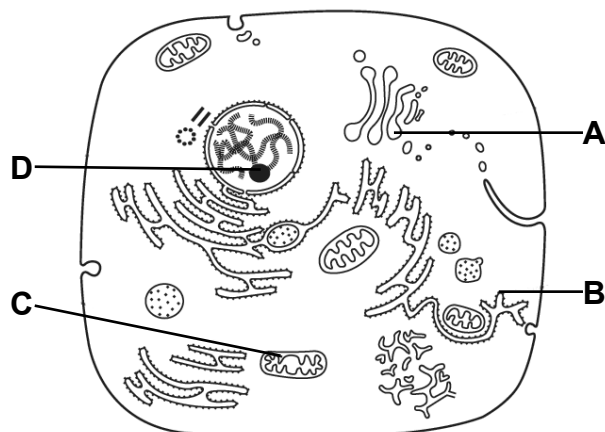
	greater in diameter	how many times
A	eukaryote	25.5
B	eukaryote	28.0
C	prokaryote	2.8
D	prokaryote	35.7

- 8 The actual diameter of a red blood cell is $300\text{ }\mu\text{m}$. The cell was magnified to a diameter of 6 cm .

What is the magnification of the cell?

- A $\times 20$
 B $\times 60$
 C $\times 200$
 D $\times 60\,000$

- 9 The diagram shows an electron micrograph of an animal cell.

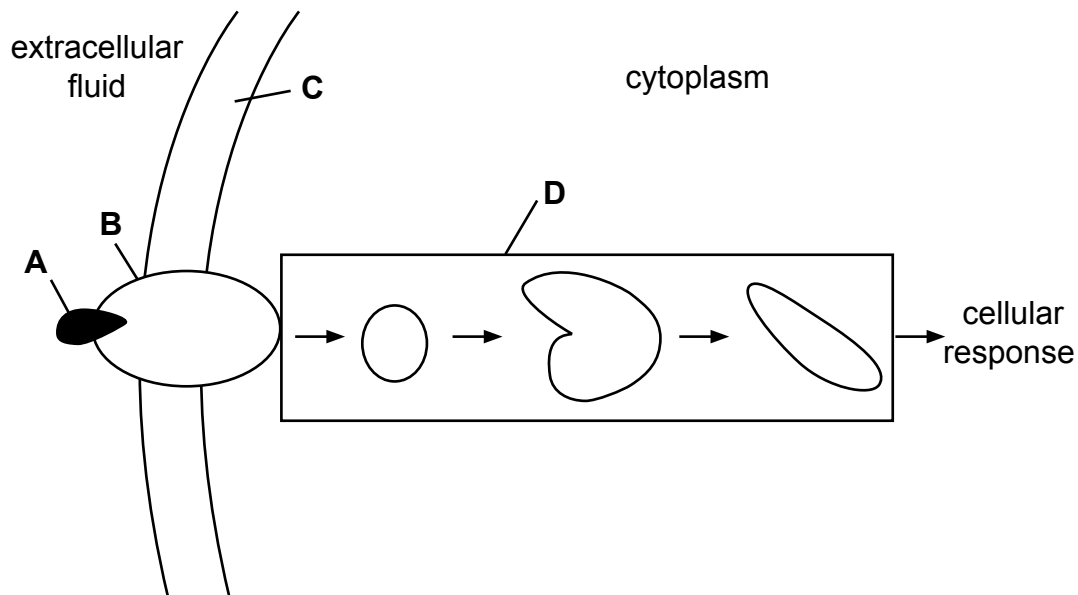


Which structure controls and regulates the activities of the cell?

- 10 Which is a feature of all prokaryotic cells?

- A absence of nucleus
- B absence of ribosomes
- C presence of chloroplasts
- D presence of mitochondria

- 11 The diagram shows the process of cell signalling.



Which letter represents the ligand molecule?

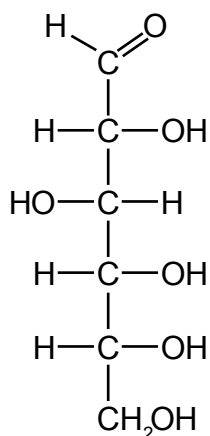
- 12 What happens as a cell increases in size?
- A its surface area increases faster than its volume
 - B its surface area remains unchanged
 - C its volume and surface area increases at the same rate
 - D its volume increases faster than its surface area
- 13 A plant cell is placed in a solution with more negative water potential than the cell contents.

Which change occurs in the cell and what causes the change?

	change	cause
A	cell becomes more flaccid	water moves out of the cell
B	cell becomes more flaccid	solution moves into the cell
C	cell becomes more turgid	water moves into the cell
D	cell becomes more turgid	solution moves out of the cell

- 14 Which term is defined as a carbohydrate whose molecule consist of a number of sugar molecules bonded together?
- A monomer
 - B monosaccharide
 - C polypeptide
 - D polysaccharide

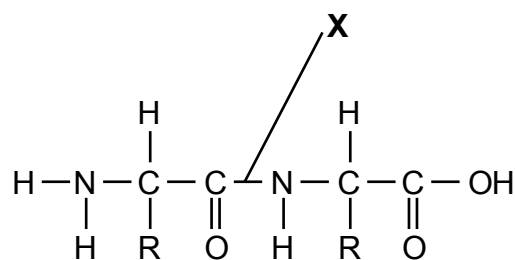
- 15 The diagram shows a structure of a molecule.



Which test on a polymer of this molecule will give a positive result?

- A adding biuret solution
- B adding iodine solution
- C heating with Benedict's solution
- D shaking with ethanol and water

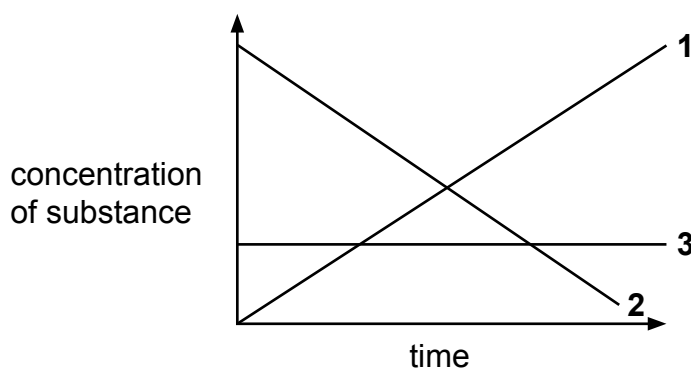
- 16** The diagram shows the molecule formed through a condensation reaction between two amino acids.



Which bond is formed at **X**?

- A** disulphide
 - B** hydrogen
 - C** peptide
 - D** ionic
- 17** What is the importance of iron in the haemoglobin molecule?
- A** it defends the body against pathogens
 - B** it ensures the blood colour remains red
 - C** it ensures the number of red to white blood cells is proportional
 - D** it binds with oxygen to aid its transports to the body tissues
- 18** Which physical property of water means a relatively large amount of energy is needed to change it from liquid state to gas?
- A** hydrogen bonding
 - B** high latent heat of vapourisation
 - C** high specific heat capacity
 - D** solvent action

- 19 The graph shows how the concentration of components 1, 2, and 3 of an enzyme-catalysed reaction changes with time.



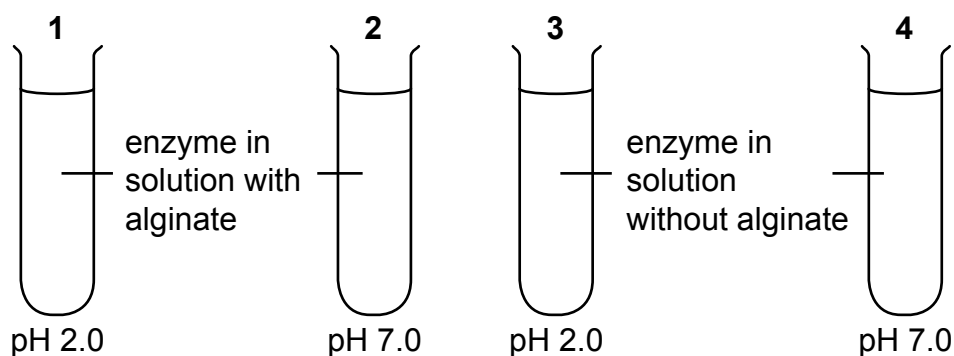
Which row identifies the components of this reaction?

	component 1	component 2	component 3
A	substrate	enzyme	product
B	substrate	product	enzyme
C	product	substrate	enzyme
D	product	enzyme	substrate

- 20 What will be the effects of a competitive inhibitor on the $V_{\max}$ and K_m in an enzyme-controlled reaction?

	$V_{\max}$	K_m
A	decreases	increases
B	increases	decreases
C	remains the same	decreases
D	remains the same	increases

- 21 A student was investigating the effect of pH on the activity of enzyme amylase. Amylase is affected by low pH when immobilised in an alginate gel. The apparatus was set up as follows:



In which test-tube(s) would the reaction be most affected by the pH?

- A** 1 only **B** 3 only **C** 1 and 3 only **D** 2 and 4 only

22 Which statement best describes the adaptations of companion cells for their roles?

- A** empty cells for allowing water to flow freely
- B** few organelles to reduce the resistance to flow
- C** many plasmodesmata for communication between cells
- D** strong lignified walls for supporting the plant

23 A student put a layer of plant epidermal cells on a microscope slide. The student put a drop of potassium nitrate solution on the layer of cells and observed that

- the cell surface membrane of many of the cells had separated from the cell wall
- the cytoplasm and cell contents had shrunk.

What explains these observations?

	direction of water movement	process taking place
A	cell to solution	active transport
B	cell to solution	osmosis
C	solution to cell	osmosis
D	solution to cell	active transport

24 Which factor does **not** directly influence mass flow in the phloem sap?

- A** concentration gradient of sucrose in the phloem
- B** concentration of starch in sink tissues
- C** hydrostatic pressure gradient in the phloem
- D** rate of carbohydrate storage in the sink tissues

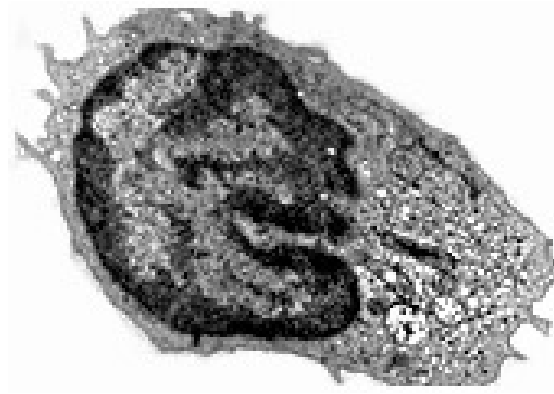
25 Which force is **not** responsible for the movement of water from the roots up through the leaves?

- A** adhesion
- B** cohesion-tension
- C** gravity
- D** transpiration pull

26 What is a function of the hairs (trichomes) on leaves of xerophytes?

- A** adding a waterproof layer
- B** covering the stomata
- C** increasing the surface area
- D** trapping a layer of moist air

27 The diagram shows an image of a blood cell as seen under an electron microscope.



Which blood cell is shown in the diagram?

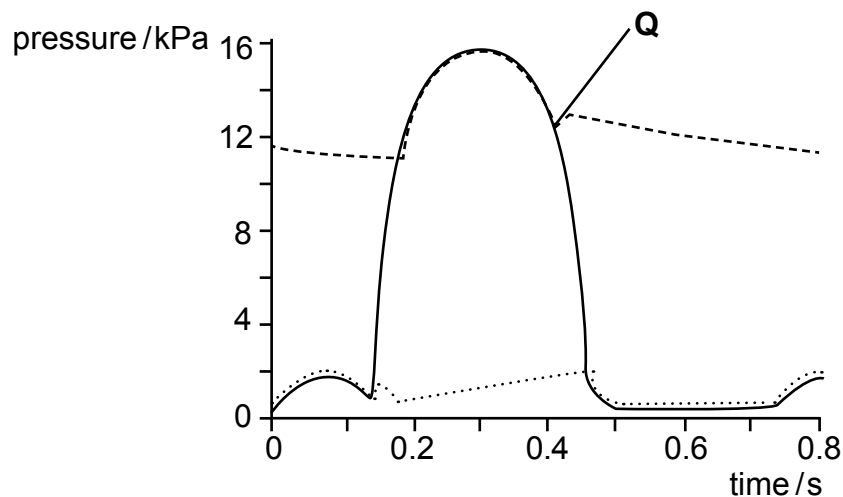
- A** lymphocyte
- B** red blood cell
- C** monocyte
- D** neutrophil

28 Which of these statements about the formation of haemoglobinic acid are correct?

- 1 It promotes the dissociation of oxygen from haemoglobin.
- 2 It removes excess hydrogen ions preventing blood becoming too acidic.
- 3 It is linked to the action of carbonic anhydrase.

- A** 1 only **B** 1 and 2 only **C** 2 and 3 only **D** 1, 2 and 3

29 The graph shows the changes in blood pressure during a cardiac cycle.



What happens at **Q**?

- A** atrio-ventricular valve closes
- B** atrio-ventricular valve opens
- C** semilunar valve closes
- D** semilunar valve opens

30 The following is a list of events of how the heart action is initiated in the cardiac cycle. They are not in the correct order.

- 1 The bundle branches conduct the impulses through the interventricular septum.
- 2 The impulses pause (0.1 sec) at the atrioventricular (AV) node.
- 3 The Purkinje tissues stimulate the contractile cells of both ventricles
- 4 The sinoatrial (SA) node generates impulses.
- 5 The atrioventricular (AV) bundle connects the atria to the ventricles.

Which row places the events in the correct order?

- A** 2 5 3 4 1
B 2 4 1 5 3
C 4 2 5 1 3
D 4 3 2 5 1

31 Which disease is a non-infectious disease?

- A** cholera
B measles
C sickle cell anaemia
D tuberculosis

32 Why is it necessary for a person with a bacterial infection to finish the course of antibiotics treatment?

- A** to increase the concentration of antibiotic slowly to a level which is lethal to the bacteria
B to maintain a concentration of antibiotic in the body which is lethal to the bacteria
C to prevent the development of resistant strains of bacteria
D to select and kill the resistant strains of bacteria

33 On which cells can the non-self antigens be found and which cells of the immune system recognise them?

	where found	cells that recognise them
A	on all body cells	phagocytes
B	on all body cells	lymphocytes
C	on foreign cells only	phagocytes
D	on foreign cells only	lymphocytes

- 34** When a particular pathogen enters the human body for the second time, the person may not show any symptoms.

What is the reason for the absence of symptoms during the second entry of the same pathogen in the body?

- A** the pathogen is no longer infectious
- B** faster response due to the presence of memory cells
- C** pathogen too weak to cause any symptoms
- D** symptoms for the same pathogen cannot be presented twice

- 35** Monoclonal antibodies are used in both diagnosis and treatment of diseases.

Which statement is an example of the role of monoclonal antibodies in the diagnosis of diseases?

- A** They are cultured to multiply in response to pathogens.
- B** They bind with cancerous cells and make them visible to lymphocytes.
- C** They detect the presence of HIV antigens.
- D** The plasma cells of a small animal is fused with cancer cells.

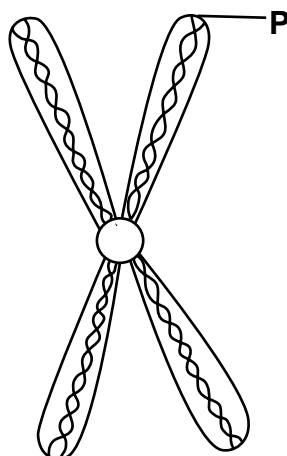
- 36** Which row correctly shows the parts of the gaseous exchange system and their functions?

	ciliated epithelium cells	goblet cells	mucous glands
A	secrete mucus into ducts	sweep mucus onto surfaces	produce mucus to cover the surfaces
B	sweep mucus onto surfaces	secrete mucus into ducts	produce mucus to cover the surfaces
C	produce mucus to cover the surfaces	secrete mucus into ducts	sweep mucus onto surfaces
D	sweep mucus onto surfaces	produce mucus to cover the surfaces	secrete mucus into ducts

- 37** Which is a short term effect of carbon monoxide on the cardiovascular system?

- A** causes blood to clot and restrict the flow
- B** causes narrowing of blood vessels
- C** reduces oxygen delivery to tissues
- D** leads to the formation of plaques in the blood vessels

38 The diagram shows a simplified structure of a chromosome.



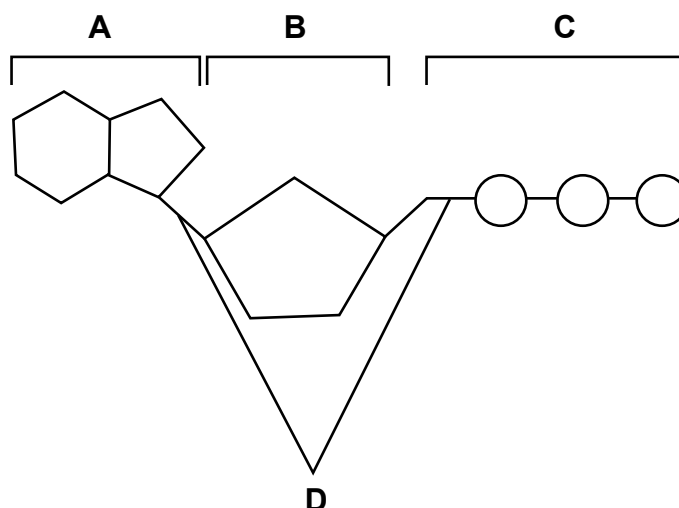
What is the name of structure **P**?

- A centromere
- B DNA
- C histone protein
- D telomere

39 Why are stem cells important in the replacement of damaged cells?

- A stem cells have gone through cell specialisation
- B they are essential for the formation of reproductive cells
- C they can produce only specialised body cells
- D they replace cells that cannot undergo mitosis

40 The diagram shows a simplified structure of a nucleotide, ATP.



Which part shows that ATP is phosphorylated?